

NISHCHAL MARUR

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SUMMARY

Machine Learning Engineer with 4+ years of experience building end-to-end production ML systems. Built vision-based verification pipelines at 96% TPR, low-latency recommendation and retrieval systems at 40ms p95, multimodal agents that reduced proposal creation from hours to under a minute, and 10x-compressed edge models for mobile deployment.

EXPERIENCE

AI Engineer Intern | Connyct Inc | New York Nov 2025 - Jan 2026

- Built a two-tower recommendation engine and hybrid RAG retrieval pipeline on AWS Lambda using Elasticsearch BM25 and dense vector kNN retrieval, achieving Recall@10 of 0.85 and <40ms p95 latency for personalized college events.
- Deployed location-aware reranking and A/B testing frameworks for an LLM orchestrator with MCP tools, directly optimizing CTR and search relevance through composite scoring and Redis caching.

Research Assistant | Perception and Robotics Group | Maryland Jun 2025 - Aug 2025

- Developed a cross-modal audio generation pipeline that converts performer pose sequences from video into violin audio using DDSP, Transformer encoders, and autoencoder-based MIDI synthesis, enabling direct audio generation from motion capture.

Machine Learning Engineer II | Entrupy Inc | Bangalore Aug 2021 - Aug 2024

- Automated 95% of manual verification for 150K+ monthly luxury sneakers and bags by building a multi-stage verification pipeline using segmentation, depth estimation, and LoFTR feature matching, achieving 96% TPR at 5% FPR.
- Achieved 10x model compression for on-device deployment via EfficientNet distillation from SAM-HQ and FP16 quantization, enabling real-time auto-capture with 96% IoU on consumer iOS devices.
- Led R&D of a 3D document unwarping system using synthetic data that outperformed SOTA benchmarks, resulting in a 23% increase in OCR accuracy with 0.84 SSIM for real-world documents.
- Reduced time-to-production for new models from weeks to days by designing synthetic data pipelines using Blender and Stable Diffusion to simulate camera intrinsics, lighting, and complex material textures.
- Reduced model drift response time from days to under 2 hours by building automated monitoring with false-positive analysis, GradCAM interpretability checks, DynamoDB-backed performance tracking, and Lambda/SQS real-time alerting.

MLOps Intern | IBM | Bangalore Jan 2021 - Jul 2021

- Reduced inference latency by 15ms in IBM Watson Cloud deployments through Go-based concurrency optimization and Kubernetes batch prediction pipelines and evaluated performance trade-offs across Tensorflow, PyTorch and ONNX runtimes.

Software Engineering Intern | SLK Software | Bangalore May 2020 - Jul 2020

- Eliminated 10+ hours of weekly debugging time by building a centralized log aggregation system using ELK Stack, Filebeat, and Node.js APIs across distributed components.

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, Scala, Bash

ML Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, ONNX, TensorRT, Scikit-Learn, Pandas, NumPy, Open3D, SAM

GenAI & Retrieval: LangChain, LangGraph, LlamaIndex, vLLM, LoRA, Elasticsearch, MCP, Pinecone, Stable Diffusion

Cloud & Infra: AWS, GCP, Kubernetes, Docker, Ray, Kubeflow, MLflow, Triton, Redis, Kafka, PostgreSQL, MongoDB, CI/CD

ML Domains: Computer Vision, Large Language Models, Model Optimization, Recommendation Systems, AI Agents, RAG, Edge AI, MLOps

PROJECTS

CAFBrain: Multimodal LLM Platform (Agentic RAG | LLM) - [GitHub](#)

- Built an agentic RAG system that reduced grant proposal creation time from hours to under a minute, using Langchain and LangGraph with multi-step tool routing across 5000+ multimodal documents using FAISS.

FSTChangeNet: Temporal Change Retrieval (Computer Vision | Multimodal) - [GitHub](#)

- Designed a dual-encoder architecture with custom frequency-spatial-temporal attention module and LoRA fine-tuned RemoteCLIP for vision-language alignment, achieving 64% Recall@10 on natural-language change queries for satellite images.

Scalable DBaaS for RideShare (Distributed Systems | Cloud Infra) - [Github](#)

- Deployed a fault-tolerant Database-as-a-Service on AWS supporting 2000+ RPS with Zookeeper-based leader election, automatic failover, and read/write routing via RabbitMQ RPC, achieving zero-downtime recovery under node failure.

EDUCATION

Master of Science (M.S) in Machine Learning Aug 2024 - May 2026

University of Maryland, College Park

GPA: 3.8/4

Bachelor of Technology (B.Tech) in Computer Science Aug 2017 - May 2021

PES University, Bangalore

GPA: 3.6/4

CERTIFICATIONS

Generative AI with LLMs (Coursera) | 5-Day Gen AI Intensive Course with Google (Kaggle) | Go: The Complete Developer's guide (Udemy)